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Applied Problems

Averages: e.g. before their final exam a student has test scores of 72, 80, 65, 78, and 60. If the final exam counts as $\frac{1}{3}$ of the total grade, what score must the student receive on the final for a total grade of 76?

$2/3$ $\qquad\qquad\qquad\qquad$ $1/3$

Test Scores	Final
72	
80	X
65	
78	
60	
Avg = 71	

$$\frac{2}{3}\cdot 71 + \frac{1}{3}\cdot x = 76 \Rightarrow 142 + x = 228$$

$$\Rightarrow \boxed{x = 86}$$

iscounts/markups: e.g. a clothing store holding a sale advertises a 20% discount on all prices. If a shirt is on sale for \$30, what was its pre-sale price.

Pre-sale Price	Sale Price
x	$x-(20\%\ \text{of}\ x)$ $=x-0.2x$ $=(1-0.2)x$ $=0.8x$
?	\$30

20% disc ↓ 20% mark ↓

$=(37.5\times 0.8)\times 1.2)$

$=30\times(1.2)$

$=36$

$\neq 37.5$

$30=0.8x$

$x=\dfrac{30}{0.8}$

$\boxed{x=\$37.5}$

Simple Interest: e.g. an investment firm has
(page 64) $100,000 to invest for a client and decides to invest it in two stocks, A and B. The expected annual simple interest for stock A is 15%, but there is some risk involved, and the client does not wish to invest more than $50,000 in this stock.

~~The annual rate of return on stock B is The an~~

The annual simple interest on stock B is 10%. Determine whether there is a way of investing the money so that the annual interest is:

a) $12,000 b) $13,000 → not possible

What is the maximum possible interest we can generate?

$I_A = 50{,}000 \times 0.15 \times 1 = 7{,}500$

$I_B = 50{,}000 \times 0.10 \times 1 = 5{,}000$

Best-case scenario

$$I_A = 7{,}500, \quad I_B = 5{,}000$$

$$I_A + I_B = 12{,}500$$

How to get 12,000?

x = investment in stock A

$$I_A = x \cdot 0.15 = 0.15x$$

$$I_B = (100{,}000 - x) \cdot 0.10$$

$$= 10{,}000 - 0.10x$$

$$I_A + I_B = 0.15x + (10{,}000 - 0.10x)$$

$$= 0.05x + 10{,}000$$

When is this 12,000?

$$0.05x + 10{,}000 = 12{,}000$$

$$0.05x = 2{,}000$$

$$x = 40{,}000$$

Mixtures: e.g. a pharmacist is to prepare 15ml of special eye drops for a glaucoma patient. The eyedrops must have a 2% active ingredient but the pharmacist only has 10% and 1% solutions in stock. How much of each type of solution should be used to prepare the prescription?

Pharmacist wants a 2% active ingredient solution of 15ml volume

$\Rightarrow$ 2% of 15ml = 15ml × 0.02
= 0.3ml

Let's say pharmacist uses x ml of 10% soln and y ml of 1% soln

ACTIVE INGR. from 10% soln of xml vol. + ACTIVE INGR. from 1% soln of yml vol. = 0.3 ~~ml~~

$0.10x + 0.01y = 0.3$

Volume of 10% soln + Volume of 1% soln = 15 ~~ml~~

$\underbrace{x + y = 15}$
$\downarrow$
rearrange for x in terms of y

$\underbrace{x = 15 - y}$
$\downarrow$
Plug into other equation

$0.10\underset{\substack{\downarrow \\ 15-y}}{x} + 0.01y = 0.3$

$0.10(15 - y) + 0.01y = 0.3$

$1.5 - 0.1y + 0.01y = 0.3$

$-0.09y = -1.2$

$\boxed{y \approx 13.3\ ml}$

$x = 15 - y$

$\approx 15 - 13.3$

$\boxed{x \approx 1.7\ ml}$

distance, speed, time: e.g: a boy can row a boat at a constant rate of 5 mi/hr in still water. He rows upstream for 15 minutes and then rows downstream returning to his starting point in 12 minutes

$d = vt$

$v = \frac{d}{t}$

a) Find the rate of the current

b) Find the total distance travelled.

15 min

$= 15 \ (1 \text{ min})$

$1 \text{ hr} = 60 \text{ mins}$

$\Rightarrow 1 \text{ min} = \frac{1}{60} \text{ hr}$

$= 15 \left(\frac{1}{60} \text{ hr}\right)$

$= \frac{15}{60} \text{ hr} = 0.25 \text{ hr}$

a) 0.56 mi/hr

b) 2.22 mi

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